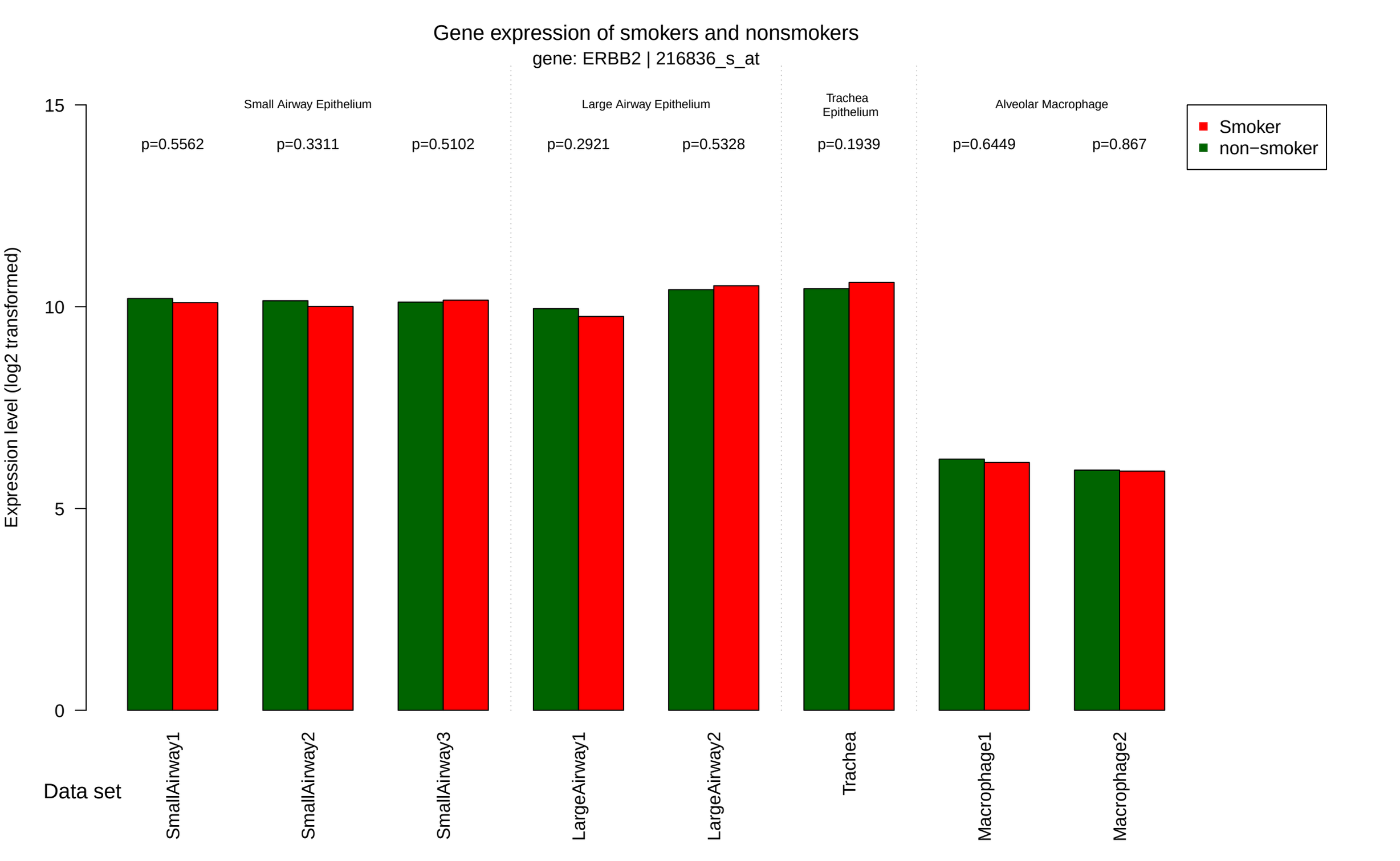
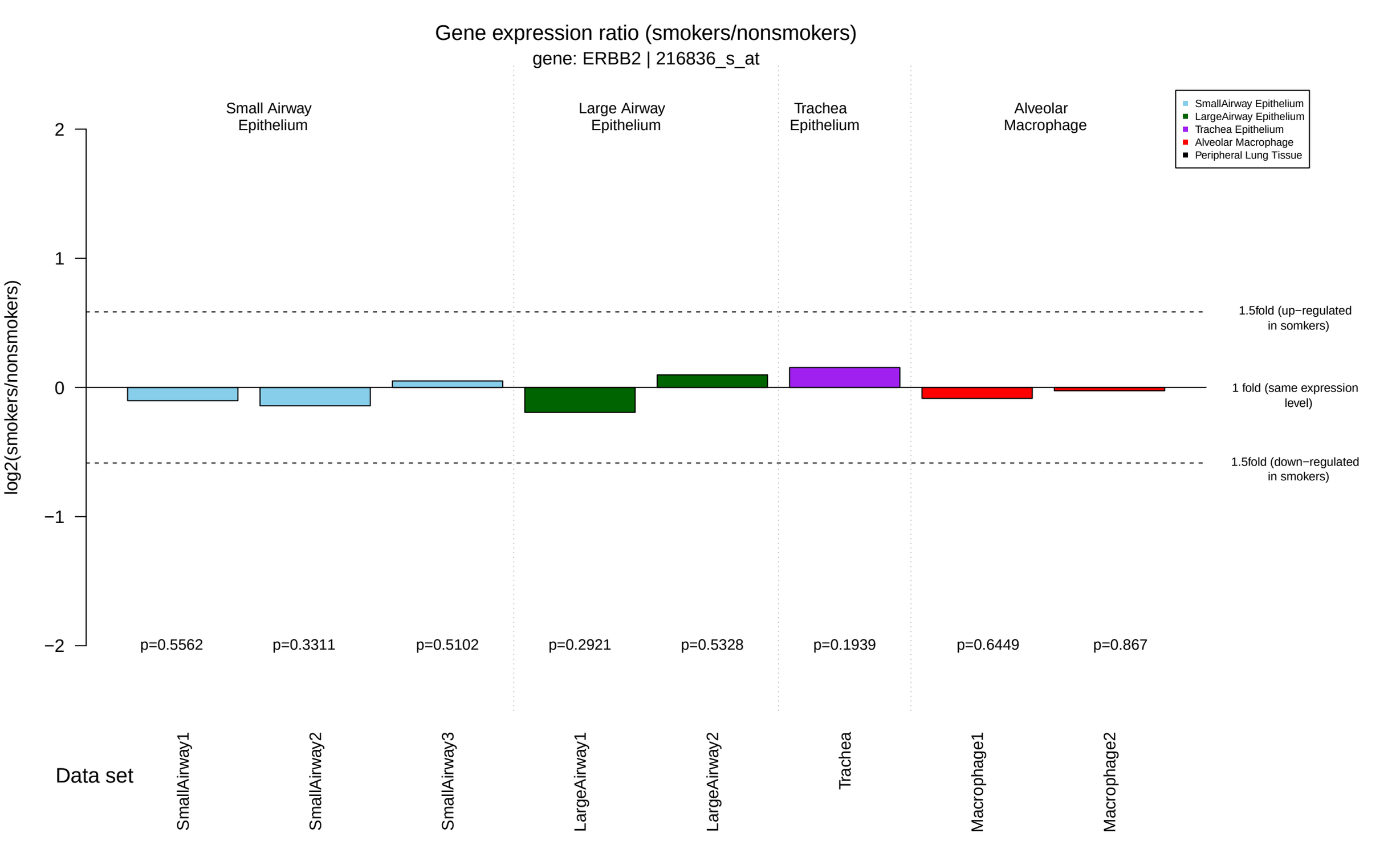
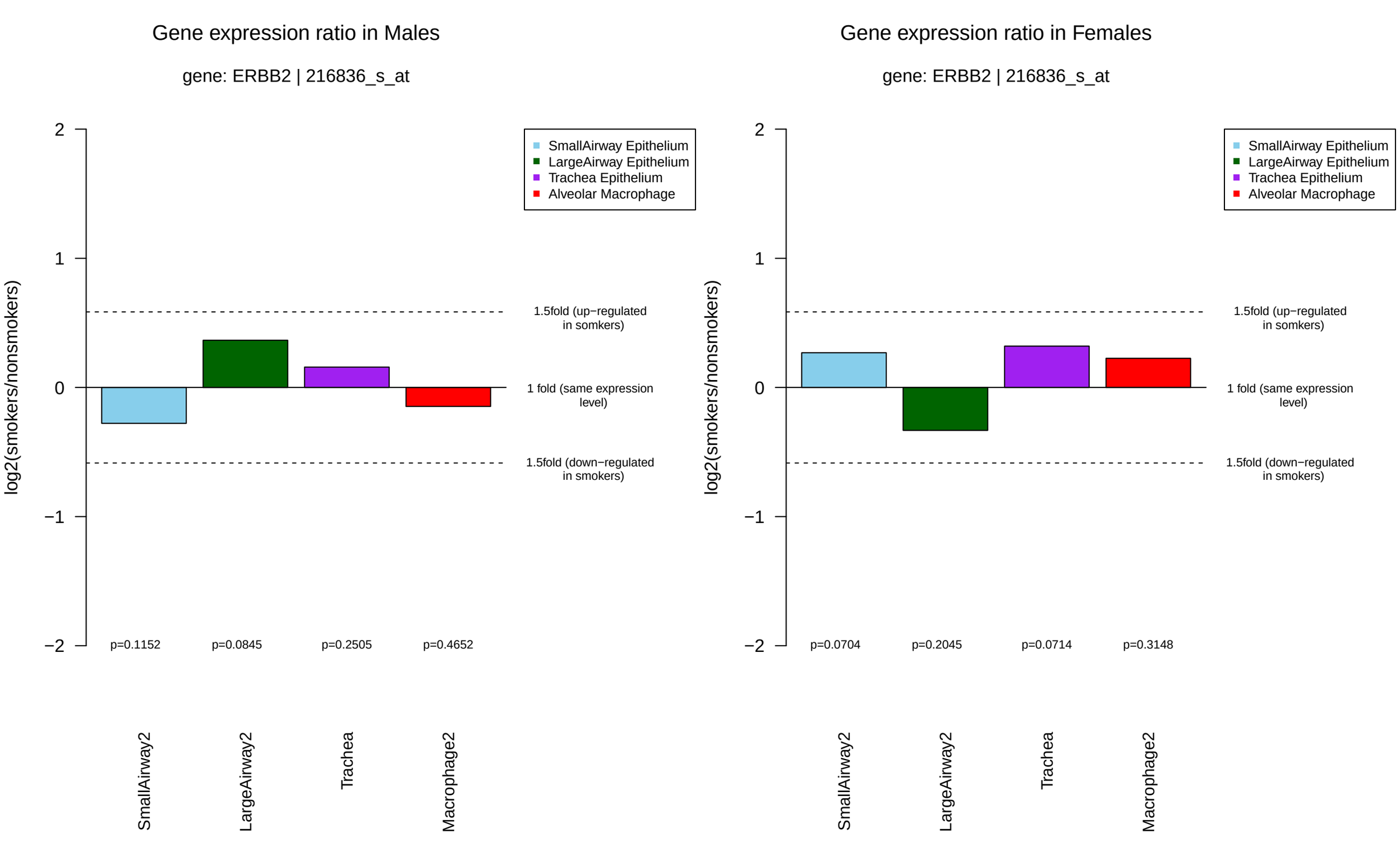
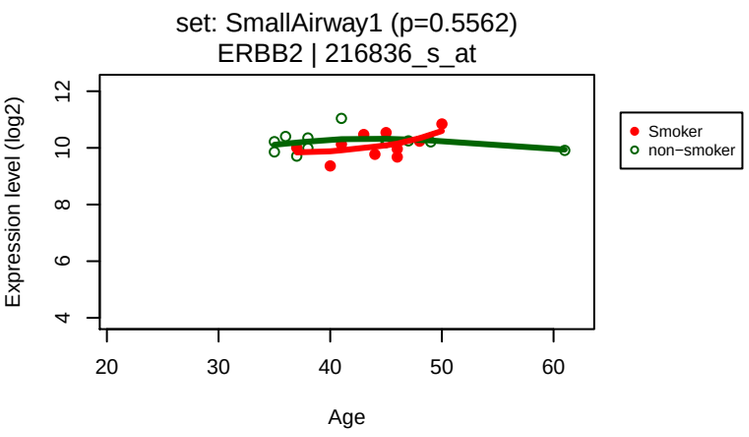
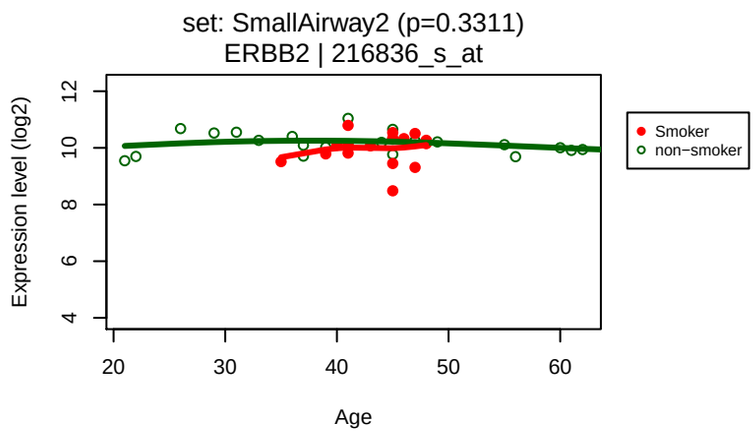
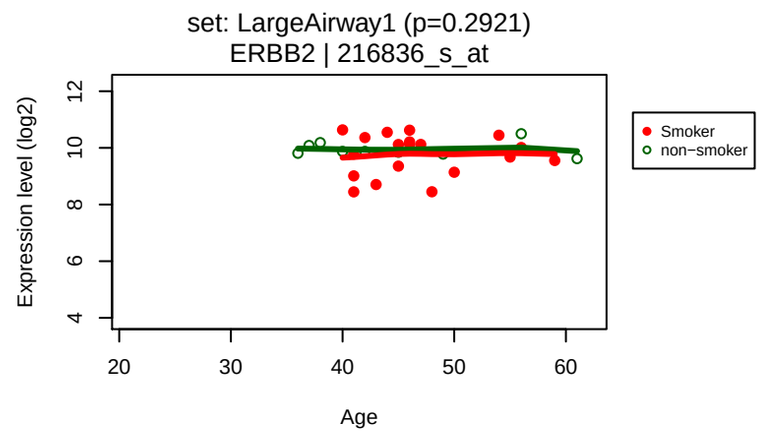

Gene expression of smokers and nonsmokers
gene: ERBB2 | 216836_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5562
p=0.3311
p=0.5102
p=0.2921
p=0.5328
p=0.1939
p=0.6449
p=0.867
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

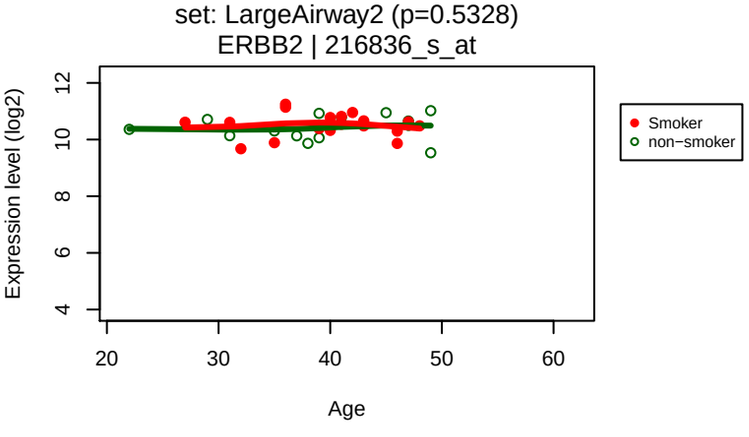

Gene expression ratio (smokers/nonsmokers)
gene: ERBB2 | 216836_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5562
p=0.3311
p=0.5102
p=0.2921
p=0.5328
p=0.1939
p=0.6449
p=0.867
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

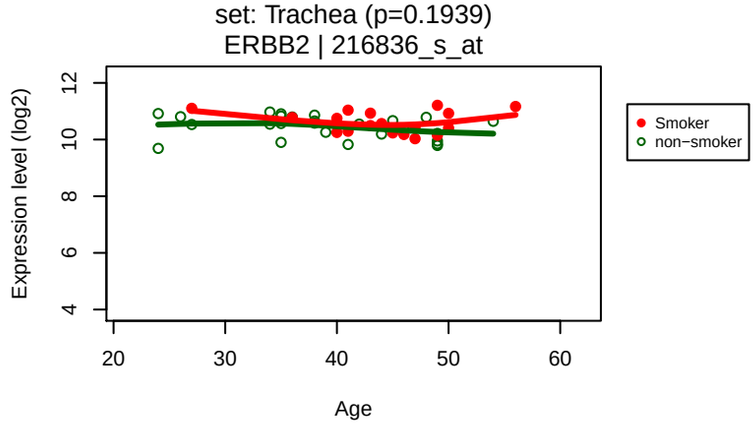

Gene expression ratio in Males
gene: ERBB2 | 216836_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1152
p=0.0845
p=0.2505
p=0.4652
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERBB2 | 216836_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0704
p=0.2045
p=0.0714
p=0.3148
SmallAirway2
LargeAirway2
Trachea
Macrophage2

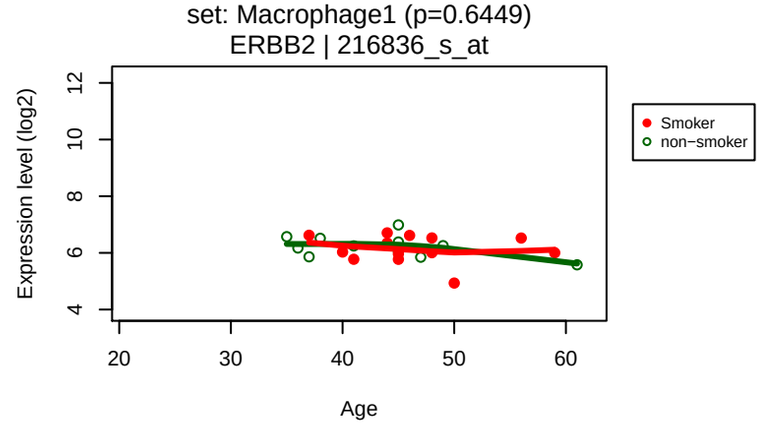

set: SmallAirway1 (p=0.5562)
ERBB2 | 216836_s_at
Expression level (log2)
Age
Smoker
non−smoker

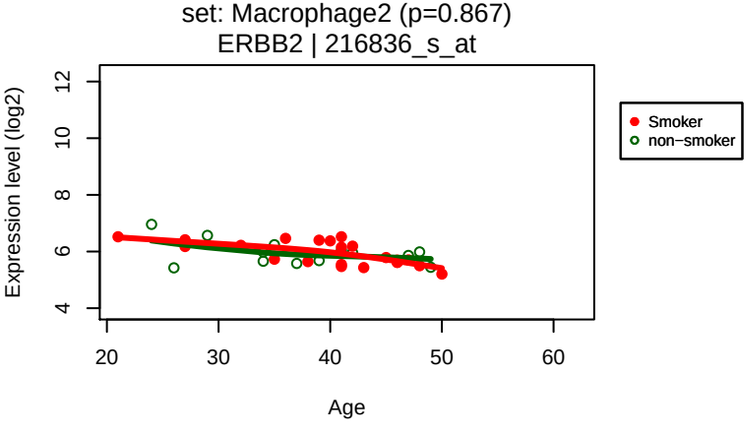

set: SmallAirway2 (p=0.3311)
ERBB2 | 216836_s_at
Expression level (log2)
Age
Smoker
non−smoker

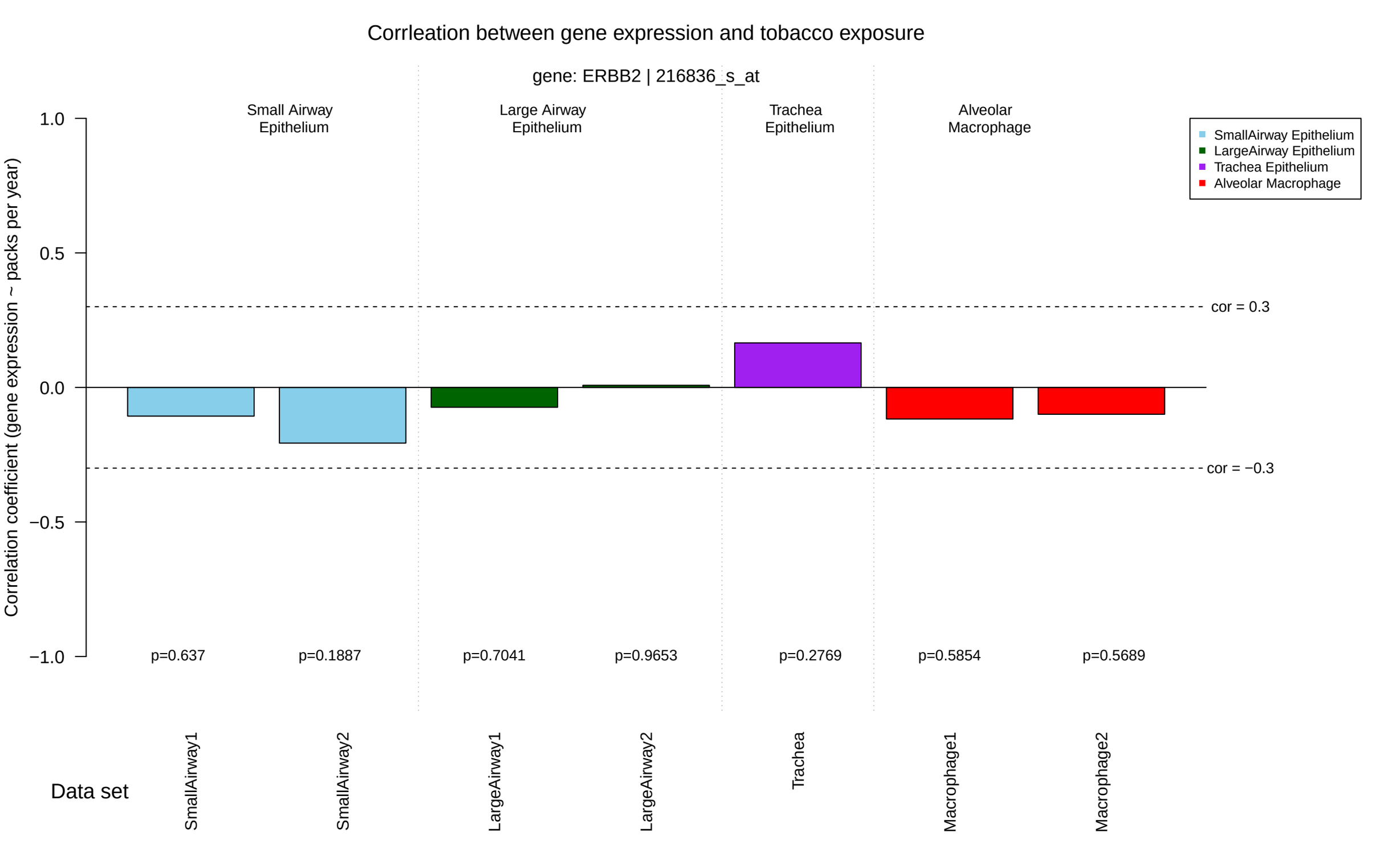

set: LargeAirway1 (p=0.2921)
ERBB2 | 216836_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5328)
ERBB2 | 216836_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1939)
ERBB2 | 216836_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6449)
ERBB2 | 216836_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.867)
ERBB2 | 216836_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERBB2 | 216836_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.637
p=0.1887
p=0.7041
p=0.9653
p=0.2769
p=0.5854
p=0.5689
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2